

CSS

Animations & Transitions

Session 1 — Detailed Lesson Guide

■ Objective	Understand CSS transitions and how to make smooth hover effects
■ Level	Beginner-friendly (Kids / Junior learners)
■ Duration	~60 minutes
■ Tools	Any code editor + browser (Chrome / Edge recommended)
■ Homework	Button that changes colour, size & shape on hover

■ Section 1 — What Is Animation?

Animation is the art of making things **MOVE, CHANGE, or come ALIVE** on a webpage — like the cartoons you see on apps and websites every day. It is not just about looks; animations guide the user's attention, show that something is happening, and make the experience feel polished and professional.

■ Real-world examples you already know:

- The **Swiggy** loading bike that drives across the screen while your food is being prepared.
- The **PhonePe green tick** that appears after a successful payment.
- **Google Doodle** letters that bounce or transform on special days.
- A **Diwali diya** flame that flickers on a festival website.
- **Flipkart** buttons that glow or grow when you hover your cursor over them.

Why does animation matter?

Without Animation	With Animation
Page feels flat and static	Page feels alive and engaging
Changes happen instantly — jarring	Changes are smooth — satisfying
Hard to notice what changed	User's eye is drawn to the right place
Looks like a plain HTML document	Looks like a professional website

■ Section 2 — The Magic of CSS

CSS stands for **Cascading Style Sheets**. It is the language that controls how your HTML looks — colours, sizes, fonts, spacing, and yes — **animations!** You do not need to draw anything by hand. A few lines of code tell the browser exactly how to move, change colour, grow, shrink, or spin.

■ Think of CSS as a Magic Wand

When you wave a magic wand, things transform instantly — or slowly, or with a sparkle. CSS is exactly like that wand for your webpage. You *describe* what you want, and the browser *does* it. The best part? You can control **how fast** the magic happens, **what changes**, and **when it starts**.

Two CSS Animation Tools — an Overview

Tool	What it does	When to use
CSS Transition	Smooth change between two states	On hover, click, or focus
CSS Animation (@keyframes)	Continuous or complex movement	Looping effects, loading spinners

■ This session focuses on **CSS Transition**. Next session covers @keyframes.

■ Section 3 — CSS Transition: The Slow-Change Tool

A **CSS Transition** lets you tell the browser: *When this element changes, take your time!* Instead of a harsh, sudden jump, the change glides smoothly from one state to another.

■ Real-life analogies for Transition:

■ Traffic Signal	Lights don't pop on and off randomly — they move Red → Yellow → Green in order, giving y
■ Sunrise Sky	The sky doesn't snap from black to blue. It slowly blends through purples, pinks, oranges —
■ Flipkart Button	When you hover over a 'Buy Now' button, the colour fades from white to orange over a fractio

■ The Transition Syntax

CSS Transition Syntax

```
/* Syntax */

transition: <property> <duration>;

/* Example */

transition: background-color 1s;

/* Multiple properties at once */

transition: background-color 1s, transform 0.5s;

/* Transition ALL properties */

transition: all 0.8s;
```

Without vs. With Transition — Side by Side

■ Without Transition

```
.box {  
background-color: orange;  
}  
.box:hover {  
background-color: green;  
}
```

Color flips instantly — like a light switch. Looks abrupt.

■ With Transition

```
.box {  
background-color: orange;  
transition: background-color 1s;  
}  
.box:hover {  
background-color: green;  
}
```

Color slides smoothly over 1 second — natural and professional.

■ Section 4 — The Three Parts of a CSS Transition

1

The Selector — `.box { }`

The selector tells CSS **which element** you want to add magic to. It can be a div, a button, an image — anything on your page. You write the element's class or tag name followed by curly braces `{ }` and put all your styling rules inside.

2

The Transition Property — `transition: background-color 1s;`

This is the **magic line**! It has two parts: (a) **what to change** (e.g., background-color, width, height, transform) and (b) **how long** the change should take (e.g., 1s = 1 second, 0.3s = 0.3 seconds). This line goes in the element's normal style — not in the `:hover` block.

3

The Hover Trigger — `.box:hover { }`

The **`:hover` pseudo-class** is the trigger. It tells CSS what should change when the user moves their mouse over the element. Without a trigger, the transition has nothing to react to. Other triggers exist too — `:focus`, `:active` — but `:hover` is the most common.

■ Complete Working Example

Full Transition Example

```
/* HTML */

<div class="box">Hover over me!</div>

/* CSS */

.box {

width: 200px;

height: 100px;

background-color: orange; /* starting colour */

border-radius: 10px;

transition: background-color 1s; /* THE MAGIC LINE */

}
```

```
.box:hover {  
  
background-color: green; /* ending colour */  
  
}
```

■ **Teacher Tip:** Ask students to copy this code into a .html file and open it in Chrome. Encourage them to hover slowly and then quickly to see the transition in action!

■ The :hover Trigger — Explained

The **:hover** pseudo-class activates whenever the user's mouse cursor moves over an element.

Think of it exactly like a **motion-sensor light** in a shopping mall:

- Walk under the sensor → light turns ON.
- Walk away → light turns OFF.

With CSS transition, this ON/OFF change happens **smoothly** instead of instantly.

When the cursor leaves the element, it also transitions *back* to the original state!

■ Section 5 — Duration: How Fast or Slow?

Duration is the number you write after the property name — it controls how long the transition takes. You write it in **seconds (s)** or **milliseconds (ms)**.

Value	Speed	Real-life Feel	Best used for...
0.2s – 0.3s	■ Very Fast	Like a quick blink	Micro-interactions, small hovers
0.5s	■ Fast	Snappy and clean	Button colour changes
1s	■ Medium	Comfortable and visible	Most hover effects
2s – 3s	■ Slow	Dramatic, like sunrise	Loading effects, special reveals
5s+	■ Very Slow	Dreamy, almost imperceptible	Background colour shifts

Duration Examples

```
/* Fast – snappy button */

.btn { transition: background-color 0.3s; }

/* Medium – comfortable card hover */

.card { transition: transform 1s; }

/* Slow – dramatic reveal */

.hero { transition: opacity 3s; }
```

■ Properties You Can Transition

Property	What changes	Example value
background-color	The fill colour	orange → green
color	Text colour	black → white
width / height	Element size	100px → 200px
font-size	Text size	16px → 24px
border-radius	Corner roundness	0px → 50%
opacity	Transparency	1 → 0.5
transform: scale()	Grow or shrink	scale(1) → scale(1.3)
transform: rotate()	Spin the element	0deg → 45deg
box-shadow	Drop shadow	none → 0 4px 20px rgba(0,0,0,0.3)

■ Section 6 — Hands-On Activity

Now it's your turn! Open your code editor, create a new file called **transitions.html**, and try the following tasks. Remember: the best way to learn CSS is to *experiment*!

1
MUST DO

Colour Change on Hover

Create a coloured box that smoothly changes its background colour when you hover over it. Use `transition: background-color 1s;` inside your `.box` style.

2
SHOULD DO

Try Three Speeds

Change the duration to 0.3s (fast), 1s (medium), and 3s (slow) one at a time. Notice how each feels different. Which feels the most natural to you?

3
STRETCH

Grow on Hover

Add `transform: scale(1.3);` inside your `.box:hover` block. Make sure your transition includes 'transform' so the growth is smooth: `transition: background-color 1s, transform 0.5s;`

■ Starter Code — Copy This!

transitions.html — Starter Template

```
<!DOCTYPE html>

<html>

<head>

<style>

.box {

width: 200px;

height: 120px;

background-color: orange;

border-radius: 12px;

margin: 60px auto;

/* ■ ADD TRANSITION HERE */
```

```
transition: background-color 1s;

}

.box:hover {

background-color: #22C55E; /* green */

/* STRETCH: add transform below */

/* transform: scale(1.3); */

}

</style>

</head>

<body>

<div class="box"></div>

</body>

</html>
```

■ Common Mistakes to Watch Out For:

- Putting **transition** inside the **:hover** block instead of the normal style — it won't animate back!
- Forgetting the **s** after the number (write **1s**, not just **1**).
- Trying to transition properties that cannot be animated (e.g., `display`, `font-family`).
- Using transition on a property that isn't actually changing — nothing will happen.

■ Section 7 — Key Parts Recap

Term	Meaning	Example
transition	The CSS property that enables smooth changes	<code>transition: background-color 1s;</code>
property	What you want to change smoothly	<code>background-color, transform, opacity</code>
duration	How long the change takes	<code>0.5s (fast) / 1s (medium) / 3s (slow)</code>
:hover	Trigger — activates when mouse moves over	<code>box:hover { background-color: green; }</code>
selector	Which HTML element to style	<code>.box { ... }</code>
transform: scale()	Resize the element	<code>transform: scale(1.3)</code> makes it 30% bigger

- CSS Transition makes changes happen **slowly and smoothly** — no more instant jumps.
- The **transition property** goes in the element's normal style block.
- **:hover** is the trigger — it fires when the mouse is over the element.
- **Duration** controls the speed — experiment with 0.3s, 1s, and 3s.
- You can transition **multiple properties** at once by separating them with commas.
- Transitions also happen **in reverse** when the mouse leaves!

■ Section 8 — What's Next?

Next Session Preview: CSS @keyframes Animations

In Session 1, you learned how to react to user actions (hover). In the next session, you will learn how to make things move **all by themselves** — no hovering needed! The secret? The powerful **@keyframes** rule.

What you'll create	CSS trick used
■ Stars twinkling in a night sky	@keyframes + animation-iteration-count: infinite
■ Diwali diya flame flickering	@keyframes opacity + scale
■ A loading spinner	@keyframes rotate
■ Elements that fade in on page load	@keyframes opacity from 0 → 1

■ Homework Challenge

Practice makes perfect! Here are your homework tasks for this session:

1**MUST: Button with Colour + Size Change**

Create a `<button>` element that changes its **background colour** AND grows slightly (use **transform: scale(1.1)**) when hovered. Use the transition property for both. Try writing it from memory before looking at your notes!

2**BONUS: Square to Circle**

Start with a square div (**border-radius: 0%**). On hover, change it to a perfect circle (**border-radius: 50%**). Make the shape change smooth with transition. Also try changing the colour at the same time!

3**SHOW OFF: Teach Your Family**

Show your family the webpage you created! Explain what CSS Transition means in your own words. Can you explain what `:hover` does? Teaching is the best way to remember!

■ Bonus Exploration (for the super curious!):

Look up **transition-timing-function**. Try values like **ease**, **ease-in**, **ease-out**, **ease-in-out**, and **linear**. How does each one feel different? Which do you like best?

Example: **transition: background-color 1s ease-in-out;**

■ You're a CSS Animator Now!

Every great website you visit uses CSS transitions just like the ones you learned today. The next time you hover over a button and see it glow or grow, you'll know exactly how it was made. **Keep experimenting — the best CSS artists are the ones who try the most things!**